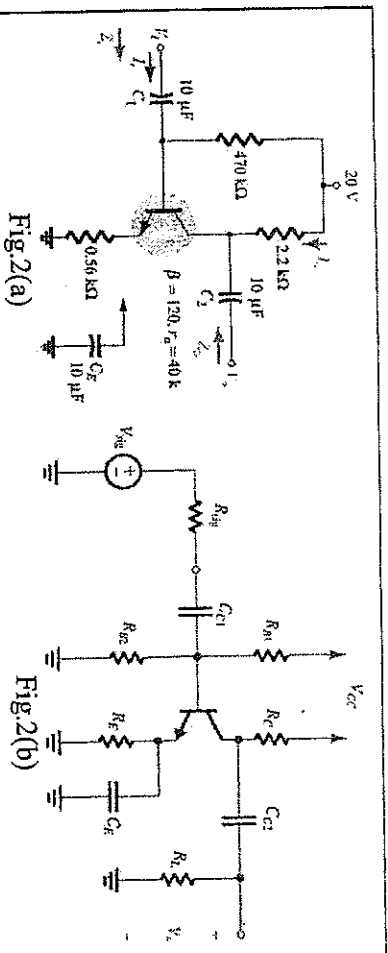
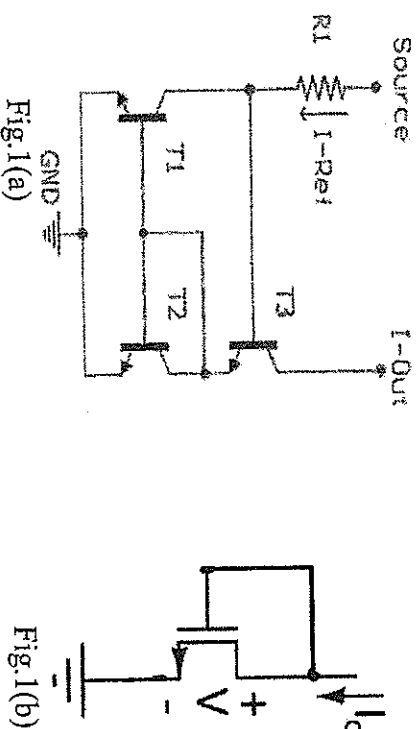
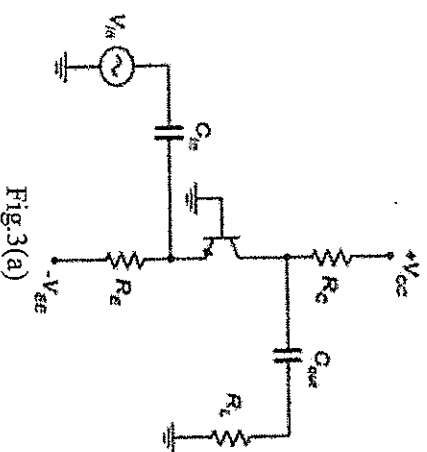
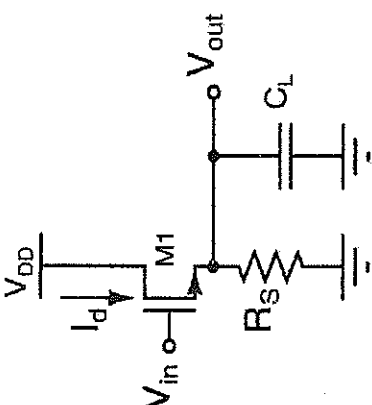
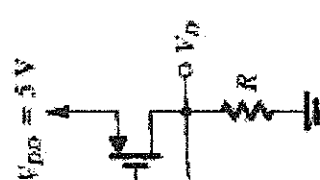
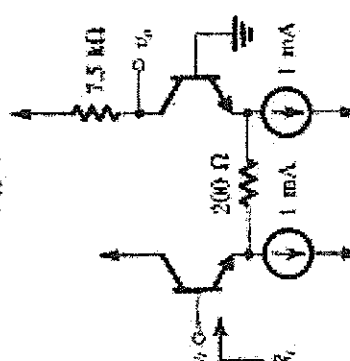


Q.no	Question	Marks	CO
1(a)	Derive an expression for current gain for the circuit shown in Fig.1(a). Assume all the transistors are matched.	2	CO1
1(b)	Show that the circuit given in Fig.1(b) is equivalent to $1/g_m$.	1	CO1
2(a)	Determine the following for emitter-bypassed CE amplifier. Refer Fig. 2(a). a) re. (b) Zi. (c) Zo. (d) Av. (e) Ai.	1	CO1
2(b)	Derive an expression for ω_{p1} for the Circuit shown in Fig.2(b) using high frequency pi-model of the BJT.	2	CO2



3(a)	Derive an expression of R_{in} , R_{out} , A_v (over all voltage gain) for the circuit given in Fig.3 (a).	2	CO1
3(b)	Explain the channel length modulation and body effect in the MOSFET.	1	CO1
4(a)	Derive the expression for overall voltage gain for the circuit shown in Fig. 4(a).	2	CO2
4(b)	Explain Miller-effect and discuss the significance in calculating dominant pole with an example.	1	CO2



	 Fig4(a)			CO2	
5(a)	The PMOS transistor shown in Fig.5(a) has $V_t = -2\text{V}$, $\mu_p C_{ox} = 8\mu\text{A}/\text{V}^2$, $L = 10\mu\text{m}$, $W = 10\mu\text{m}$ and $V_D = 7\text{V}$. Find the value of I_D and R.			1	CO2
5(b)	Find the voltage gain and input resistance of the amplifier given in Fig.5(b). assume $\beta = 100$.			2	CO2
	 Fig5(a)			 Fig.5(b)	